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## **M 204      CONDESATE PUMPS**

### **1.      SCOPE OF WORK**

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2 “Summary of Work”**.

Three (3) x 50% duty capacity Vertical Condensate Pumps and appurtenances, including motor drives, minimum flow facilities etc., as per this Specification. All the pumps shall be identical.

### **2.      CODES AND STANDARDS**

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 “Codes and Standards”**.

The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5 “Codes and Standards”** or international equivalent standards are specifically applicable to the design, manufacture, and testing of the equipment included in this specifications

|      |                                                  |
|------|--------------------------------------------------|
| AISI | American Iron and Steel Institute                |
| ANSI | American National Standards Institute.           |
| ASME | American Society of Mechanical Engineers.        |
| ASTM | American Society of Testing and Materials.       |
| BS   | British Standard                                 |
| DIN  | German Institute for Standardization             |
| EN   | European Committee for Standardization           |
| IEC  | International Electrotechnical Commission.       |
| IEEE | Institute of Electrical and Electronic Engineers |
| JIS  | Japan Industrial Standard                        |

### 3. DESIGN AND OPERATING CONDITIONS

Three (3) Vertical Condensate Pumps with a capacity of 50% each will supply the condensate from the condenser hotwell trough strainer and isolating valves to the deaerator.

The pumps will be located indoor and mounted vertically in the turbine room grade floor near the condenser.

The pumps shall be heavy-duty type suitable for continuous service.

Pumps shall be designed for the following conditions:

- (a). Pumps suction .....Condensate from condenser hotwell through an isolating valve
- (b). Pump discharge.....Through a gland steam condenser, piping, valves, etc.into the deaerating heater.
- (c). Pump design capacity,

each pump..... Bidder shall determine total condensate flow based on the plant operating conditions:

Pumps shall also be sized to prevent high water level in the condenser during turbine bypass operation as specified in **Part 4, Section 3, M-202 “Condenser”**.

- (d). Condenser shell absolute  
pressure..... Bidder to determine.

- (e). Normal temperature  
of water ..... Saturated at condenser pressure.

- (f). Condensate .....pH 8 - 10

- (g). Net positive suction  
head required and  
available .....Bidder to determine.

- (h). Total pump head ..... Bidder to determine based on the static head differential pressure between condenser and deaerator and the dynamic pressure losses in the supply line . A margin of 10% shall be added to the calculated head of the pump.
- (i). The proposed pump head at shut-off shall be 15 to 25% greater than the head at the design point.
- (j). The pump NPSH required shall be 25% lower than the available NPSH at all points on the pump curve. If the manufacture requires a higher margin, this margin shall be applied.
- (k). Attenuating flow from the condensate pumps will be required for the Turbine Bypass Systems. The flow required shall be adequate to cool the Steam flow to the temperature acceptable to the turbine exhaust, the Condenser, etc.
- (l). For environmental considerations see **Part 4, Subsection 2.16**

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## **4. DETAILED REQUIREMENTS**

### **4.1 Pumps**

Pumps shall be of single or multiple stage, vertical, centrifugal type design suitable for the intended service.

Pump suction and discharge connections shall be flanged. Flanged connections shall conform to ANSI Standards.

The pump and motor shafts shall be aligned and connected through a suitable coupling. Pump and motor shafts shall be keyed to the coupling.

The vertical driving motor shall be mounted on a substantial base. The driving motor shall be of the full voltage starting type with solid shaft and all parts of the equipment shall be designed to withstand resulting stresses. The pump and motor design shall be such that no damage will result from reverse flow due to check valve failure in the discharge line.

Bidder shall be responsible for the pump and motor combination operating in dynamic balance without undue vibration or noise. The critical speed of the complete unit (pump and motor) shall be at least 25% above or below the operating speed.

Each pump shall be designed so that the entire pump unit may be removed from the can for maintenance and repair. Lifting lugs or eyes shall be provided as required to facilitate handling.

A vent connection shall be supplied at the high point of the pump casing on the suction side to operate as a vent for the casing. This connection for each pump shall be individually piped to the condenser.

Pumps shall have non-overloading characteristics and shall have a stable head-capacity characteristic so that the head shall continue to increase with a decreasing capacity until a maximum head is reached at zero capacity.

Each pump shall operate satisfactorily without distress in any part through the full capacity range from recirculation flow to the maximum specified capacity.

## 4.2 Materials

Bidder shall be responsible for providing pumps of design and materials best suited for the specified service condition. If in the opinion of the Bidder the any specified materials are not adequate, Bidder shall, in the proposal, recommend those materials which he deemed most suitable for the service.

Material selection shall be in accordance with the following:

|                            |                                                                      |       |
|----------------------------|----------------------------------------------------------------------|-------|
| Shaft.....                 | Stainless steel                                                      |       |
| Shaft sleeves .....        | Hardened stainless steel ground smooth under the packing and bearing |       |
| Pump bowl material.. ..... | Cast iron                                                            | CDA   |
| Impellers .....            | Stainless steel                                                      | M-002 |

## 4.3 Shafts and Impellers

The pump shafts shall be supported by bearings above and below each impeller. Shaft whipping shall be prevented by guide bearings.

The impellers shall be statically and dynamically balanced radially to assure smooth operation under all conditions

Shaft sleeves shall be provided at the stuffing boxes and expected wear points to protect the shafts.

#### **4.4 Wear Rings and Seals**

All wear rings, interstage seals or other parts (either rotating or stationary) forming the pump bowl assembly shall be securely held and locked in place. Such attachments shall be liberally designed and renewable.

#### **4.5 Bearings**

Bearings shall be of such design that they will operate over a long period of years with a minimum of wear and replacement.

Thrust bearings shall be similar to the Kingsbury or Michell bearing type or equivalent. Roller type thrust bearings may be provided where appropriate.

#### **4.6 Lubrication**

The pumps shall be lubricated by the condensate being pumped. No external lubrication shall be required.

#### **4.7 Pump Support Bases**

The pumps shall be provided with the mounting bases suitable for grouting and bolting to the foundation. Bidder shall provide leveling screws in the mounting base and grout the base as required.

The base shall be provided with large openings in the sides for easy access to the shaft coupling and seal or stuffing box.

#### **4.8 Pressure Parts**

Manufacturer shall meet all applicable requirements of Section VIII, Division 1, of the ASME Boiler and Pressure Vessel Code for calculating the thickness of pressure retaining parts and in sizing the cover bolting.

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#### **4.9 Painting**

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter. All unmachined exterior ferrous

surfaces shall painted in accordance with **Part 4, Section3, Technical Requirements S-420 and S-430.**

#### **4.10 Shipment**

Inlet and outlet of the pumps shall be plugged and blanked for shipment.

### **5. PERFORMANCE TESTING AND GUARANTEES**

#### **5.1 Shop Test**

Each pump shall be shop tested by the Bidder as follows:

- (a). A factory performance test of total head, efficiency, brake power and NPSH required versus flow shall be performed on each pump in accordance with HIS requirements. During testing the pump shall operate at design speed and with 38°C maximum water temperature.
- (b). All tests shall have a minimum of five flow points including the guaranteed minimum flow, guaranteed design flow and run-out flow (125 % of design flow if not otherwise specified). Bidder shall submit test curves to the Owner for approval prior to pump shipment.
- (c). Brake power shall be determined preferably by use of a calibrated torque measuring device.
- (d). Test Curves
  - The As-Tested performance:
    - a. Tolerances for the as tested pump capacity, power, efficiency at the design point shall be per HIS requirements.
    - b. Shutoff head shall be within plus or minus 10 percent of the preliminary pump curve shut-off head.
  - NPSH Test
    - Presence of Cavitation shall be based on a 1 % drop of head across the first stage impeller.
- (e). Vibration Readings during the performance test upper bearing housing (flexible coupling) (rigid coupling) shall be taken on the pump and at the top motor bearing. Two readings 90° apart shall be taken at each location in the horizontal direction.
- (f). Vibration readings shall not exceed the limits specified in **Part 4, section**

**3, M-014.**

- (g). All parts of the pump which will operate under pressure shall be given a hydrostatic test of 1.5 times the design pressure. Parts normally operating under vacuum shall be given a hydrostatic test of 3.5 bar(g) minimum.

**5.2 Site Test**

After equipment is put in service a running test shall be conducted to ascertain that the equipment is performing properly in every respect. Any deficiencies brought out by this test shall be promptly corrected by Bidder.

**6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)**

Quality assurance/control shall conform to the requirements of **Part 4, Subsection 2.7.**

**7. INSTRUMENTATION AND CONTROLS**

**7.1 General**

The Bidder shall be responsible for the design, supply and installation of instrumentation and control devices as specified herein and for the complete design, supply and installation of all local instrumentation and control system, devices, test instrument and control panels integral with the condensate pumps whether or not they are specified herein. The detail of Instrumentation and Control device refer to the **Part 4 Section 3, I-001.**

**7.2 Local Control Panel**

The Bidder shall furnish and install a local control panel for the pumps.

The local control panel will only be used for emergency, maintenance and testing purposes, and shall contain all control and monitoring devices which do not need to be installed at a local measuring location, and all operator interfaces such as indicators, push buttons, lights, control switches and alarms. For details of Local Control Panel requirement refer to **Part 4, Section 3, I-002.**

## **8. ELECTRICAL SPECIFICATIONS**

### **8.1 General**

The Bidder shall furnish and install all electrical equipment and accessories required for the condensate pumps.

For general electrical requirements, see **Part 4, Section 3, E-001**.

### **8.2 Motors**

For detailed electrical requirements, see **Part 4, Section 3, E-210** for Low Voltage Motors.

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